OFFICIAL LINKS

🌐 **Website**
https://braintube.site

📢 **WhatsApp Channel (VTU Notes & Updates)**
https://wa.me/918884624741

▶️ **YouTube Channel**
https://youtube.com/@braintube-v9w?si

BMATE101 – MATHEMATICS-I (EEE)

MODULE–2 : SERIES EXPANSION & MULTIVARIABLE CALCULUS

TAYLOR & MACLAURIN SERIES, LIMITS

1. Taylor's Series (Statement & Use)

If a function $f(x)$is sufficiently differentiable at $x = a$, then its Taylor series expansion about $x = a$is:

$$\boxed{f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots}$$

Purpose:
Used to approximate functions near a given point and simplify computations.

2. Maclaurin Series

Maclaurin series is a special case of Taylor series about $x = 0$.

$$\boxed{f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots}$$

3. Standard Maclaurin Expansions

These are **very important** for problems:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

4. Problems on Series Expansion

Problem 1

Expand e^xup to x^3.

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$$

Problem 2

Expand $\sin x$up to x^5.

$$\sin\ x \approx x - \frac{x^3}{6} + \frac{x^5}{120}$$

5. Indeterminate Forms

An expression is **indeterminate** if its value is not directly obtainable.

Common forms:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 1^{\infty}$$

These are evaluated using **L'Hospital's Rule**.

6. L'Hospital's Rule

If:

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

Then:

$$\boxed{\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}}$$

7. Problems Using L'Hospital's Rule

Problem 3

Evaluate:

$$\lim_{x\to 0} \frac{\sin x}{x}$$
$$= \lim_{x\to 0} \frac{\cos x}{1} = 1$$

Problem 4

Evaluate:

$$\lim_{x\to 0} \frac{e^x - 1}{x}$$
$$= \lim_{x\to 0} \frac{e^x}{1} = 1$$

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8. Applications of Series & Limits

- Signal approximation
- Error estimation
- Circuit analysis
- Control systems
- Numerical computation

PARTIAL DIFFERENTIATION, TOTAL DERIVATIVE, JACOBIANS, MAXIMA & MINIMA

9. Functions of Several Variables

A function involving more than one independent variable is called a **function of several variables**.

Example:

$$z = f(x, y) = x^2 + y^2$$

10. Partial Differentiation

Definition

The partial derivative of a function with respect to one variable is obtained by **treating the other variables as constants**.

First-order Partial Derivatives

If $z = f(x, y)$:

$$\boxed{\frac{\partial z}{\partial x} = f_x, \frac{\partial z}{\partial y} = f_y}$$

Example 1

If:

$$z = x^2y + 3xy^2$$
$$\frac{\partial z}{\partial x} = 2xy + 3y^2$$
$$\frac{\partial z}{\partial y} = x^2 + 6xy$$

11. Higher-Order Partial Derivatives

Second-order derivatives:

$$\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x\,\partial y}$$

For continuous functions:

$$\boxed{\frac{\partial^2 z}{\partial x\,\partial y} = \frac{\partial^2 z}{\partial y\,\partial x}}$$

12. Total Derivative

Definition

If $z = f(x, y)$where $x$and $y$depend on t, then the **total derivative** of $z$with respect to $t$is:

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}}$$

Example 2

Let:

$$z = x^2 y, x = t, y = t^2$$
$$\frac{dz}{dt} = (2xy)\frac{dx}{dt} + (x^2)\frac{dy}{dt}$$

Substitute:

$$= 2(t)(t^2)(1) + (t^2)(2t) = 2t^3 + 2t^3 = 4t^3$$

13. Jacobian

Definition

The **Jacobian** of $u(x, y)$and $v(x, y)$with respect to $x, y$is:

$$\boxed{\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}}$$

Example 3

Let:

$$u = x + y, v = x - y$$
$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$$

14. Maxima and Minima of Functions of Two Variables

Critical Points

For $z = f(x, y)$, critical points occur when:

$$f_x = 0, f_y = 0$$

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Second-Derivative Test

Define:

$$D = f_{xx}f_{yy} - (f_{xy})^2$$

- If $D > 0$and $f_{xx} > 0$→ **Minimum**
- If $D > 0$and $f_{xx} < 0$→ **Maximum**
- If $D < 0$→ **Saddle point**
- If $D = 0$→ Test inconclusive

Example 4

Find extrema of:

$$\begin{aligned}
&z = x^2 + y^2 + 2x + 4y \\
&f_x = 2x + 2 = 0 \Rightarrow x = -1 \\
&f_y = 2y + 4 = 0 \Rightarrow y = -2 \\
&f_{xx} = 2, f_{yy} = 2, f_{xy} = 0 \\
&D = (2)(2) - 0 = 4 > 0
\end{aligned}$$

Hence, **minimum** at $(-1, -2)$.

15. Applications of Multivariable Calculus

- Signal expansion
- Error analysis
- Optimization problems
- Electrical field modeling
- Control systems

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